



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

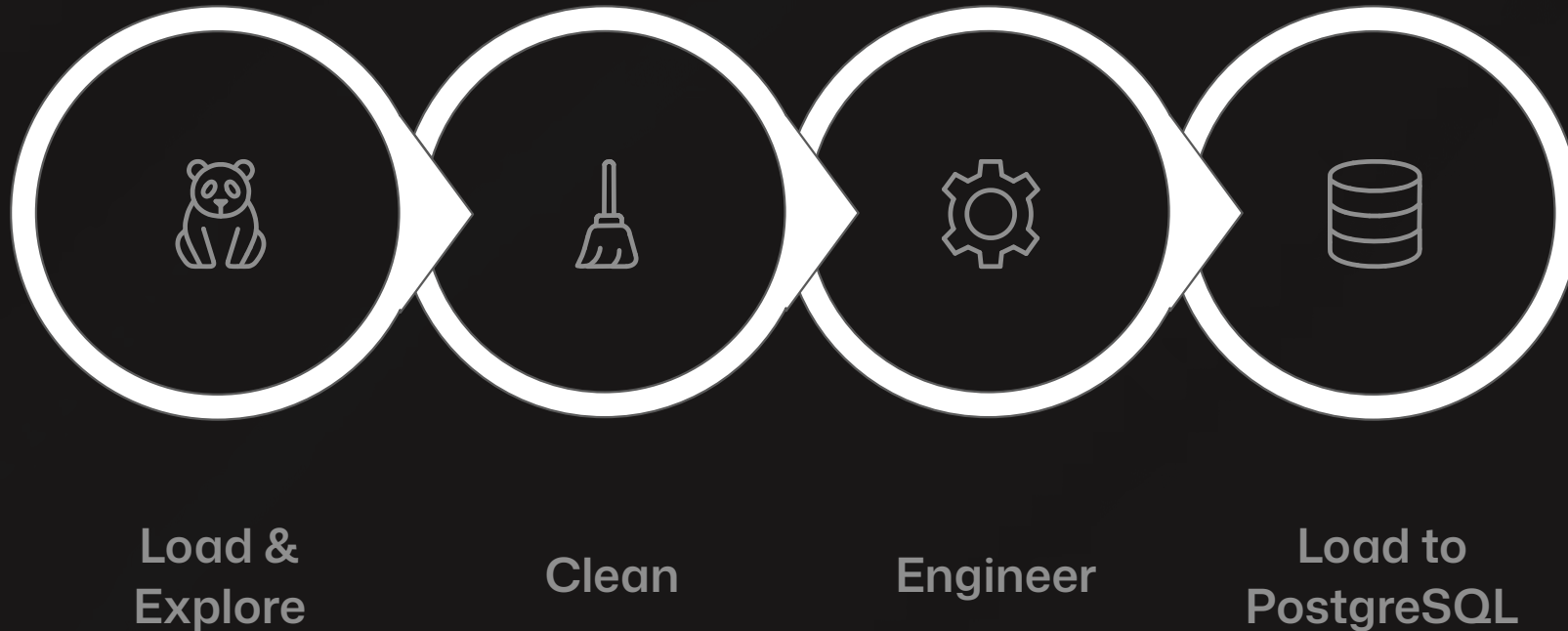
Review Rating only

\$59.78

Avg Purchase

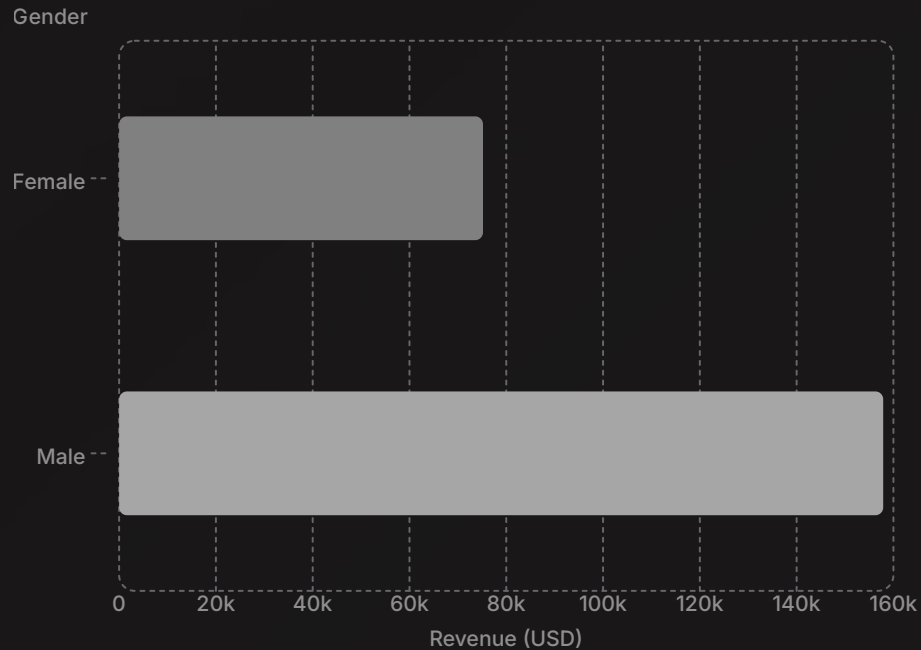


Data Preparation in Python



Snake_case standardization, redundant column removal, and database integration completed the pipeline.

Revenue by Gender



Key Insight

Male customers generated **2.1x more revenue** than female customers across 3,900 transactions.

Male

\$157,890

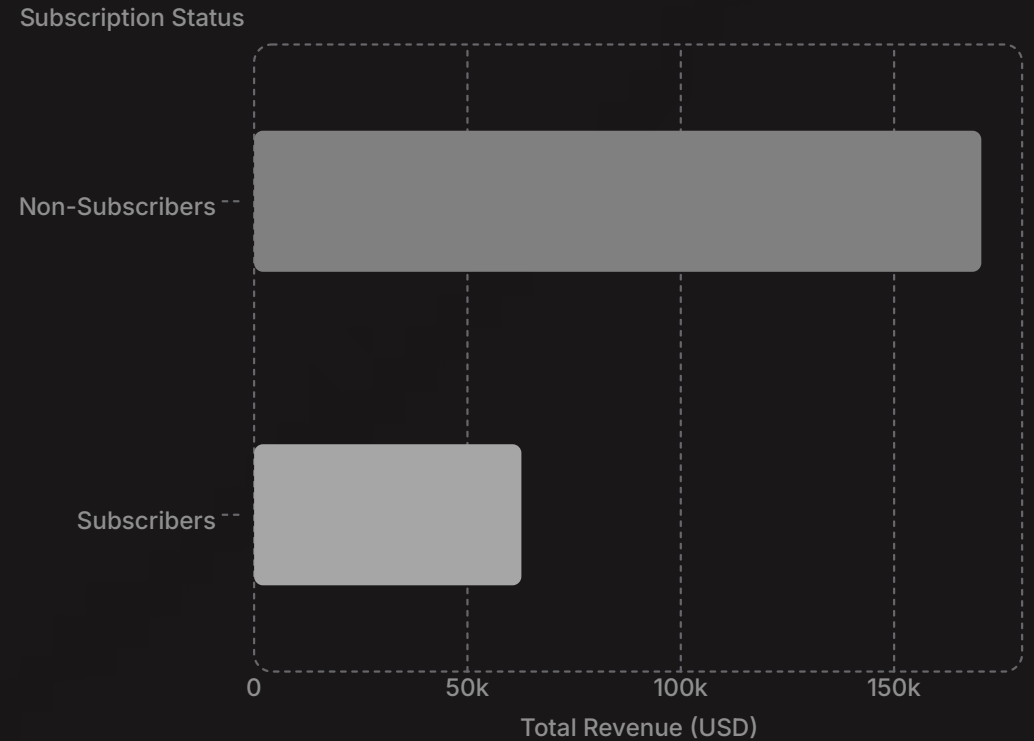
Female

\$75,191

Subscribers vs. Non-Subscribers

Revenue Share

Non-subscribers drive **73%** of customers but similar avg spend (~\$59). Subscription conversion is the growth lever.



Top Products by Rating & Discount Dependency

★ Highest Rated

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

🏷️ Most Discounted

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%



Customer Segmentation

Loyal

3,116 customers

Returning

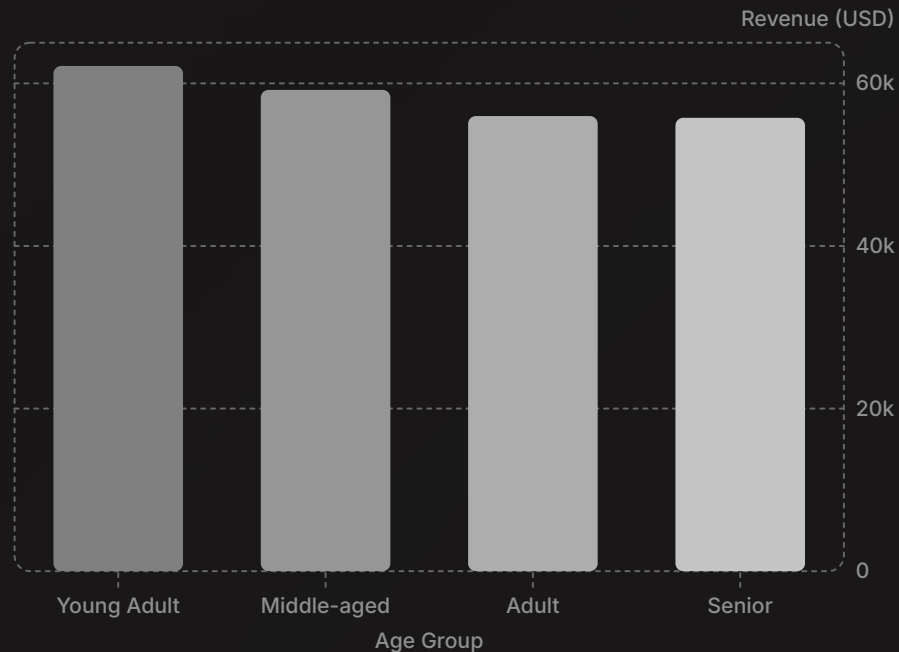
701 customers

New

83 customers

839 customers used discounts yet spent above average. 958 repeat buyers (>5 purchases) are subscribed.

Revenue by Age Group



Age Group Performance

Young Adults lead revenue at **\$62K**. Revenue declines gradually with age — a key targeting signal.

Express Shipping

Avg \$60.48 vs Standard \$58.46

Top Category

Clothing: \$104K revenue, 3.4M sales



Power BI Dashboard



Interactive Filters

Subscription, Gender, Category, Shipping Type



Key Metrics

Avg Purchase \$59.76 · Rating 3.75 · 4K Customers



Visualizations

Sales & Revenue by Category and Age Group

Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits

→ Loyalty Programs

Reward repeat buyers → Loyal segment

→ Review Discount Policy

Balance sales lift with margin control

→ Targeted Marketing

Focus on Young Adults & express-shipping users

